Computational Physics: Materials Science

Exercise 8

Equation of state of a Lennard–Jones gas

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Equation of state of a Lennard–Jones gas

The goal of this exercise is to measure the *pressure* of the Lennard–Jones (LJ) molecular dynamics (MD) simulation built on sheet 6 and to use it to map out the *equation of state* (EOS) of the LJ gas — the relation between pressure P and density ρ. The pressure is obtained from the instantaneous *virial*; the EOS is then compared, density by density, against the low-density approximation $\beta P/\rho \approx 1 + B_2\rho$, where B_2 is the second virial coefficient. The sheet is worked in three parts: (a) compute B_2 from the Mayer-f integral, plot B_2/σ^3 against ε/k_BT and interpret its sign; (b) implement the pressure estimator and report the block-averaged $\langle P\rangle$ at the reference density with a statistical error; and (c) repeat the simulation from $\rho = 0.05\,\sigma^{-3}$ up to $0.50\,\sigma^{-3}$ and compare the measured pressure to the virial EOS. The Atomic Simulation Environment (ASE) drives the dynamics. An AI assistant (Claude) was used to develop the virial/pressure kernel, the Mayer-f integration, the block-averaging and the plotting; the choice of methods, their interpretation and this report are my own.

Implementation

Code base so far and what was added

The code base inherited from sheet 6 is a periodic LJ fluid integrated with ASE. Its core is a single `numba`-compiled pair kernel (with a numerically identical vectorised NumPy fallback) that loops over all pairs inside a cubic box, applies the *minimum-image convention* and a *shifted* cutoff at $r_{\text{cut}} = 2.5\,\sigma$ (the pair energy is shifted so that $V(r_{\text{cut}}) = 0$), and returns energy and forces. The potential is wrapped as a subclass of `ase.calculators.Calculator`, the fluid is initialised on a simple-cubic grid with a Maxwell–Boltzmann velocity distribution (centre-of-mass motion removed), and the system is propagated with a Langevin NVT thermostat.

For this exercise the following was *added*:

- **Virial accumulation.** The same pair kernel now also accumulates the scalar virial $\vartheta = \sum_{i<j} \mathbf{r}_{ij} \cdot \mathbf{f}_{ij}$ in the existing pair loop (no extra cost). A per-pair sign check confirms that the attractive (s^6) term lowers the pressure and the repulsive (s^{12}) term raises it.
- **Pressure estimator.** A function returns the instantaneous pressure $P = (2K + \vartheta)/(3V)$ (Eq. (1)) in bar, using K from ASE and the explicit conversion $1\,\text{eV}/\text{Å}^3 = 1.602\,176\,634 \times 10^6\,\text{bar}$.
- **Second virial coefficient.** The Mayer-f integral (Eq. (2)) is evaluated numerically for the *same shifted-cutoff* potential, so B_2 and the simulation are consistent.
- **Statistics.** A block-averaging routine returns $\langle P\rangle$ and its standard error, plus a scan of the error versus the number of blocks to check that it has plateaued.
- **Density sweep.** A driver repeats equilibration and production over a list of reduced densities, averaging over several random seeds.

Pressure and virial

The instantaneous pressure follows the virial theorem,

$$P = \frac{1}{3V}\big(2K + \vartheta\big), \qquad \vartheta = \sum_i \mathbf{r}_i \cdot \mathbf{f}_i = \sum_i \sum_{j>i} \mathbf{r}_{ij} \cdot \mathbf{f}_{ij}, \tag{1}$$

with $V = L^3$ the (constant) box volume and K the kinetic energy. The two forms of the virial are equivalent; the pairwise form is used because the pair force $\mathbf{f}_{ij}$ is already available in the kernel, and the same minimum-image displacement $\mathbf{r}_{ij}$ that enters the force is reused for the virial. The first term, $2K/3V$, is the ideal-gas (kinetic) pressure $\rho k_B T$; the virial term carries the entire effect of the interactions. The production runs use the Langevin NVT thermostat throughout: the pressure is a *static* thermodynamic average and is therefore not biased by the momentum damping (unlike a transport quantity such as the diffusion coefficient of sheet 6), while NVT holds $T = 300\,\mathrm{K}$ most stably for a clean $\langle P \rangle$.

Second virial coefficient

The truncated virial EOS and the second virial coefficient are

$$\frac{\beta P}{\rho} \approx 1 + B_2\,\rho, \qquad B_2 = -\frac{1}{2}\int \mathrm{d}^3 r\,\left(e^{-\beta U(r)} - 1\right) = -2\pi \int_0^{r_{\mathrm{cut}}} \left(e^{-\beta U(r)} - 1\right) r^2\,\mathrm{d}r, \tag{2}$$

with $\beta = 1/k_B T$. Equation (2) is the leading correction of the *virial expansion* $\beta P/\rho = 1 + B_2\rho + B_3\rho^2 + \dots$, a density expansion of the EOS in which B_2 collects all *two-body* correlations through the Mayer function $f(r) = e^{-\beta U(r)} - 1$; the isotropy of the pair potential reduces the angular part of the integral to the factor 4π (hence the -2π prefactor). Because U enters only through $\beta U = 4\,(\varepsilon/k_B T)\,[(\sigma/r)^{12} - (\sigma/r)^6 - \mathrm{shift}]$, the dimensionless quantity B_2/σ^3 is a function of the single reduced variable $\varepsilon/k_B T$ alone. The integral is evaluated for the shifted-cutoff potential, so the integrand vanishes identically beyond r_{cut} and matches the potential actually sampled in the MD.

Simulation parameters

N	n_{eq}	n_{prod}	m [g/mol]	T [K]	ε	σ [nm]	Δt [fs]	ρ [σ^{-3}]
125	30000	30000	39.95	300	$0.3\,k_B T$	0.188	2	0.05

Table 1: Simulation parameters (Table 1 of the sheet). With $T = 300\,\mathrm{K}$ the well depth is $\varepsilon = 0.3\,k_B T \approx 7.76\,\mathrm{meV}$, the size is $\sigma = 1.88\,\text{Å}$ and the shifted cutoff is $r_{\mathrm{cut}} = 2.5\,\sigma = 4.70\,\text{Å}$, giving the reduced state point $\varepsilon/k_B T = 0.300$ (equivalently $T^* = k_B T/\varepsilon = 3.33$). The Langevin thermostat uses $\gamma = 0.01\,\mathrm{fs}^{-1}$ ($\tau = 100\,\mathrm{fs}$); pressures are sampled every 10 steps during production and block-averaged.

Results

(a) Second virial coefficient and the sign of B_2

Figure 1 shows B_2/σ^3 as a function of $\varepsilon/k_B T$. It starts positive in the high-temperature (small $\varepsilon/k_B T$) limit, decreases monotonically, crosses zero, and becomes increasingly negative as the temperature falls. At our state point, $\varepsilon/k_BT = 0.300$, the coefficient is

$$B_2/\sigma^3 = +0.277 \ > \ 0, \tag{3}$$

and the sign change (the *Boyle point*, $B_2 = 0$) occurs at $\varepsilon/k_BT \approx 0.356$, i.e. at reduced temperature $T^*_{\mathrm{B}} \approx 2.81$.

What the sign means. B_2 measures whether the *net* two-body interaction is effectively repulsive or attractive at a given temperature. When $B_2 > 0$ the configurational integral of the Mayer function is dominated by the repulsive core — particles effectively exclude volume — so the gas is *harder* to compress than an ideal gas and P lies *above* the ideal value. When $B_2 < 0$ the attractive well dominates: particles spend enough time near contact that the attraction pulls the pressure *below* ideal. At the Boyle temperature the two contributions cancel and the gas behaves ideally to first order in density. Our point sits above the Boyle temperature ($T^* = 3.33 > T^*_{\mathrm{B}} = 2.81$), where thermal energy outweighs the well depth, so repulsion wins and $B_2 > 0$ — consistent with the positive value in Fig. 1.

Expected scaling with ρ. With B_2 fixed, Eq. (2) predicts that the compressibility factor $Z \equiv \beta P/\rho$ rises *linearly* in ρ with slope B_2 (slope $B_2/\sigma^3 = +0.277$ when plotted against the reduced density $\rho^* = \rho\sigma^3$). Equivalently the pressure itself is $P = \rho k_BT\,(1 + B_2\rho) = \rho k_BT + B_2 k_BT\,\rho^2$: an ideal-gas term linear in ρ plus a positive quadratic correction. Because $B_2 > 0$ the leading deviation pushes P slightly *above* the ideal gas, and grows quadratically with density.

Range of validity. The two-term form is only the first correction of a density expansion; it is valid in the *dilute* regime where the next term $B_3\rho^2$ is negligible, i.e. at low ρ^*. As the density increases, three- and higher-body correlations become important and the truncated EOS must break down — which is exactly what part (c) demonstrates.

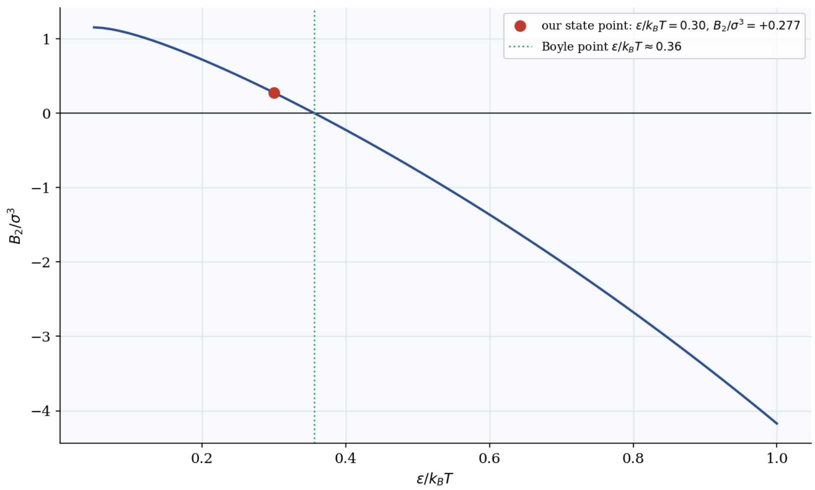


Figure 1: (a) Reduced second virial coefficient B_2/σ^3 versus ε/k_BT for the shifted-cutoff LJ potential. The coefficient is positive at high temperature (repulsion-dominated), changes sign at the Boyle point $\varepsilon/k_BT \approx 0.356$ (dotted), and turns negative at low temperature (attraction-dominated). The marker shows our state point, $\varepsilon/k_BT = 0.30$, $B_2/\sigma^3 = +0.277$.

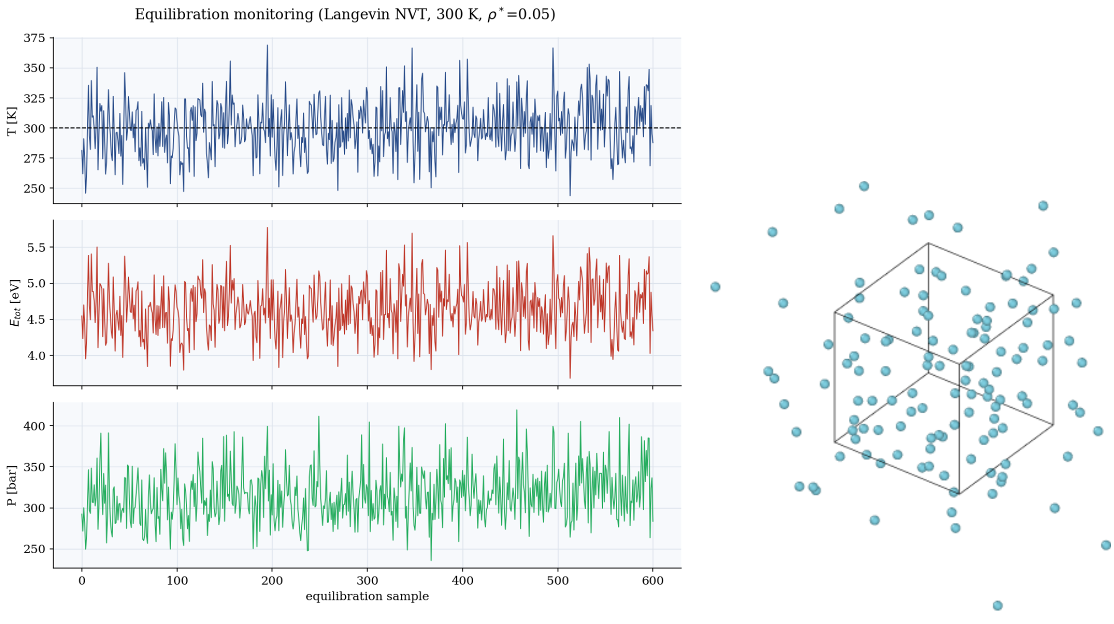
(b) Pressure measurement and block averaging

The pressure estimator of Eq. (1) was sampled every 10 steps over the production run at the reference density $\rho^* = 0.05$. Figure 2a monitors the equilibration: temperature, total energy and pressure all settle onto stationary fluctuating plateaus, with the temperature distributed about the target 300 K (measured $\langle T \rangle = 299.8$ K, $\sigma_T = 22.3$ K for the 125-particle system), confirming that the macroscopic observables are well converged before averaging. Block averaging over the production trajectory gives

$$\langle P \rangle = 317.08(211)\,\text{bar} \qquad (\rho^* = 0.05,\ 20\ \text{blocks}). \tag{4}$$

Why block averaging. Successive MD samples are *time-correlated*: the configuration at one step is close to the next, so treating the individual samples as independent would badly *underestimate* the error. Block averaging groups the series into a few long blocks; each block mean averages over many correlation times and the *block means* are approximately independent, so their spread, SEM $=$ std(block means)$/\sqrt{M}$, is an honest statistical error. A scan of the error against the number of blocks plateaus (SEM $\approx$ 1.5–2.6 bar over the range tried), confirming that the blocks are long enough to be decorrelated and that the quoted ± 2.11 bar is not an artefact of the block count.

Figure 2b is an OVITO snapshot of the same $\rho^* = 0.05$ system (125 atoms in a 25.5 Å cubic cell). The particles are sparse and disordered, as expected for a dilute gas at this state point, and stay well distributed across the periodic box — a visual confirmation that the simulation is running sensibly.



(a) Equilibration monitoring at $\rho^* = 0.05$. (b) OVITO snapshot, $\rho^* = 0.05$.

Figure 2: (b) Left: temperature, total energy and instantaneous pressure during the Langevin-NVT run at $\rho^* = 0.05$; all three fluctuate around stationary values (T about the dashed 300 K target). Right: a snapshot of the 125-atom dilute gas in its 25.5 Å cubic box.

(c) Equation of state: simulation versus the virial prediction

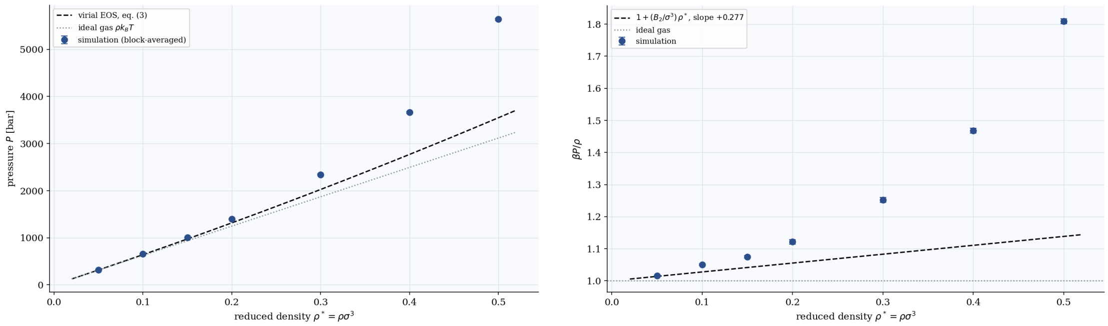
The simulation was repeated at seven densities from $\rho^* = 0.05$ to 0.50, averaging over three random seeds at each density. Table 2 lists the box length, the measured pressure and the measured compressibility factor $Z = \beta P/\rho$ alongside the two-term prediction $1 + (B_2/\sigma^3)\rho^*$. Figure 3 plots the pressure (left) and the compressibility factor (right) against density.

At the two lowest densities the simulation agrees well with the virial EOS: at $\rho^* = 0.05$ the measured $Z = 1.016$ sits essentially on the predicted 1.014, and the measured pressure lies just above the ideal gas, exactly as the positive B_2 requires. As the density grows the simulation points climb *above* the straight virial line, gently at first and then steeply: by $\rho^* = 0.30$ the measured $Z = 1.25$ already far exceeds the predicted 1.083, and by $\rho^* = 0.50$ the gap is large ($Z = 1.81$ measured versus 1.14 predicted). The $\beta P/\rho$ panel makes this clearest: the data follow the predicted slope only near the origin and then curve upward away from the line.

This is the expected behaviour. The two-term EOS keeps only the second virial coefficient, so it can only describe the dilute limit where pairwise correlations dominate. As the density rises, higher-order terms ($B_3\rho^2$ and beyond) — the contribution of three-body and many-body configurations — become significant and *add* to the pressure, which is why the measured points rise above the truncated prediction rather than below it. In practice the linear $\beta P/\rho$ law is trustworthy here only up to $\rho^* \sim 0.1$; beyond that the simulation, which contains all orders, is the reliable EOS while Eq. (2) systematically underestimates P.

ρ^* [σ^{-3}]	L [Å]	$\langle P\rangle$ [bar]	$Z = \beta P/\rho$ (sim)	Z from Eq. (2)	deviation
0.05	25.516	316.7	1.016	1.014	+0.2%
0.10	20.252	654.7	1.050	1.028	+2.2%
0.15	17.691	1004.5	1.074	1.042	+3.1%
0.20	16.074	1398.0	1.121	1.055	+6.2%
0.30	14.042	2341.6	1.252	1.083	+15.6%
0.40	12.758	3661.0	1.468	1.111	+32.2%
0.50	11.843	5639.5	1.809	1.138	+59.0%

Table 2: (c) Equation of state from the density sweep ($N = 125$, $T = 300\,\mathrm{K}$, three seeds per point). $\langle P\rangle$ is the block-averaged pressure; $Z = \beta P/\rho$ is the measured compressibility factor; the next column is the two-term prediction $1+(B_2/\sigma^3)\rho^*$ with $B_2/\sigma^3 = +0.277$. The deviation is small in the dilute limit and grows rapidly once the truncated virial expansion ceases to be valid.



(a) Pressure P versus ρ^*.

(b) Compressibility factor $\beta P/\rho$ versus ρ^*.

Figure 3: (c) Measured EOS (block-averaged points) versus the two-term virial prediction (dashed) and the ideal gas (dotted). (a) Pressure in bar. (b) Compressibility factor: the data follow the predicted slope $B_2/\sigma^3 = +0.277$ only at low density and then curve upward as higher-order virial terms become important. The truncated EOS holds up to $\rho^* \sim 0.1$ and underestimates P thereafter.

Questions to think about

- **Barostat.** A barostat couples the box volume to a target pressure, just as a thermostat couples the velocities to a target temperature: the measured P from Eq. (1) is compared to the set point and the box is rescaled to drive P towards it, moving the simulation from NVT to (approximately) NPT. The simplest is a *Berendsen* barostat, which relaxes the volume exponentially towards the target; it is robust for equilibration but does not sample the true NPT ensemble (its volume fluctuations are wrong). Proper NPT requires an extended-system barostat such as *Nosé–Hoover* (Andersen/MTK) or, when the cell shape must also change (e.g. solid–solid transitions, anisotropic stress), *Parrinello–Rahman*. With a barostat one can obtain the equilibrium density at a given pressure — the natural way to trace an EOS or locate phase transitions — rather than fixing the volume by hand as done here.
- **Sign of B_2.** As discussed in part (a): $B_2 > 0$ means the repulsive core dominates the two-body average (excluded volume, P above ideal); $B_2 < 0$ means the attractive well dominates (P below ideal); $B_2 = 0$ is the Boyle point where they balance and the gas is ideal to first order in ρ.
- **Block averaging.** Consecutive pressure samples are correlated in time, so a naive standard error treats correlated data as independent and underestimates the uncertainty. Block averaging restores independence at the level of block means and yields an honest error bar, as confirmed by the plateau of the error-versus-block-count scan.

Summary

- **Pressure estimator.** The sheet-6 LJ kernel was extended to accumulate the virial $\vartheta = \sum_{i<j} \mathbf{r}_{ij} \cdot \mathbf{f}_{ij}$ (minimum image), giving the instantaneous pressure $P = (2K + \vartheta)/(3V)$, converted to bar.
- **Second virial coefficient (a).** From the Mayer-f integral of the shifted-cutoff potential, $B_2/\sigma^3 = +0.277$ at $\varepsilon/k_BT = 0.300$, with the Boyle point at $\varepsilon/k_BT \approx 0.356$ ($T^* \approx 2.81$). A positive B_2 signals repulsion-dominated behaviour and a pressure above ideal; the leading correction scales as ρ in $\beta P/\rho$ (as ρ^2 in P), and is valid only

at low density.

- **Average pressure (b).** At $\rho^* = 0.05$, $\langle P \rangle = 317.08(211)$ bar (block averaging, 20 blocks), with $\langle T \rangle = 299.8$ K confirming the thermostat. Block averaging is required because the samples are time-correlated.
- **Equation of state (c).** Over $\rho^* = 0.05$–0.50 the measured pressure rises from ≈ 317 to ≈ 5640 bar. The two-term virial EOS matches the data only in the dilute limit ($\rho^* \lesssim 0.1$) and is increasingly exceeded at higher density, where higher-order virial terms become important — the expected breakdown of the truncated expansion.